



Sam Lu (CHAO-CHUNG LU)



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Master of UK, Brunel Uni. Major in mechanical engineering. Engaged in the manufacturing industry and specialize in NPI and supply chain development and management. Interested in coding and pursuing junior backend engineer.

Projects and Technologies Applied

Forum

SCH

basic forum functions: login, like, follow, feeds, run-time ranking

Redis cache - Reduce API response time from 595ms to 49ms.

Use dynamic attribute queries to reduce database operations and server-to-database transmissions.

- Full Stack app
- Express (Node.js)
- Postman
- MySQL CRUD
- Facebook, Google OAuth2
- Mocha test
- AWS deployment
- RDS database
- codepipeline CI/CD
- Redis for cache
- RESTful API & GraphQL API

Coffee Talk

SCH

basic twitter functions: post tweet, reply, follow, feeds, suggesting

Frontend and backend separation for team collaboration project

Due to cost considerations, the database will be migrated from AWS MySQL to Render PostgreSQL.

- Frontend-Backend separated app
- 4-person group project
- communicate by notion and postman
- React; Express (I am backend)
- PostgreSQL CRUD
- sequelize
- deploy at Github Page and Render.com
- GitHub CI/CD
- Customized login pages

Ledger & POS

SCH

Two basic applications running on the same Azure VM

Docker to package apps for different environments

Ansible to configure the vm environment

Nginx for assign ports for 2 apps

- mongoDB
- Ansible
- Nginx
- GitLab CI/CD
- Docker
- Azure VM

Card Game

- Static Website + DOM

Work Experience

PLUM INC (USA) 2018 Jun.- 2022 Oct. | Reliability Engineer & Site Manager

1. Lead supplier audits and get Bosch's order and investment by robust supplier chain design, completed PPAP, coherent FMEA and PFMEA.
2. Reduce cost at NPI stage 15% by defining reasonable component specifications and qualified by customized tests.
3. Design test plan, set up a local test dept. to Increase YPR(First Pass Yield) from 60% to 90%.
4. Second and local parts sources development to reduce around 8% cost after first MP stage.
5. Summarized CM and supplier feedback and transfer into design changes to remove NG risks.

SIRTEC INTERNATIONAL 2016 Sep.- 2017 Dec. | Supplier Development Engineer and SQE

1. Sourcing of mechanical and electrical parts like: LCD, touch panel, connector and cable, motors, hinge, insulating foam, package, power supply, PCBA, coating, pressure tank.
2. Establish test team, venue planning, define equipment spec and purchasing, test plan design, test method and documentation, training.
3. DFM to increase FPY and comply with UL, FCC/IC, CE requirements DVT stage.
4. Technician representative during local field service in the USA.